

# Multivariable Regression

## Part 1 - Categorical Predictors

Ryan Miller

# Introduction

Previously we learned about **simple linear regression** as a way of relating a quantitative explanatory variable with a quantitative outcome:

$$\text{Theoretical Model: } y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

$$\text{Fitted Model: } \hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

We extend this framework to find *adjusted effects* that account for *simultaneous changes* that occur in other explanatory variables:

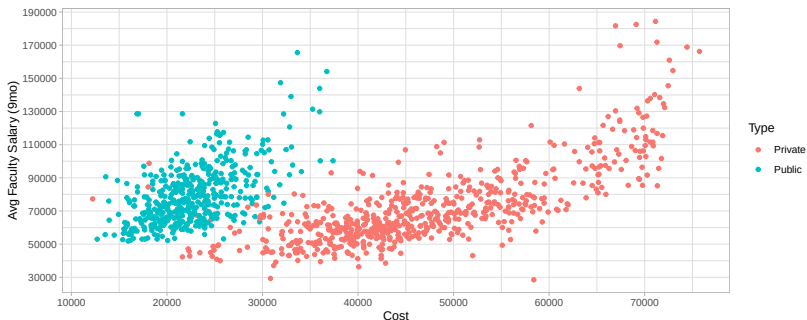
$$\text{Theoretical Model: } y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \epsilon_i$$

$$\text{Fitted Model: } \hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2} + \dots + \hat{\beta}_p x_{ip}$$

In **multivariable linear regression**, we propose that the outcome is a combination of explanatory variables and

## Example

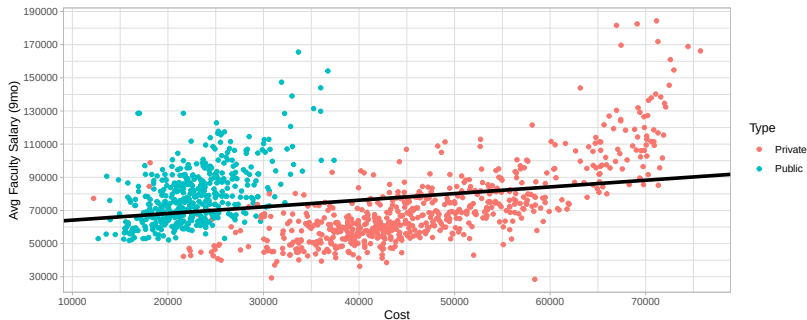
Consider the relationship between two explanatory variables, type of institution and cost, and a college's average 9-month faculty salary:



Do both explanatory variables appear related to the outcome? Do both explanatory variables appear related to each other?

## Example (cont.)

If we consider cost (simple linear regression) we get the fitted equation  $\widehat{\text{Salary}} = 60150 + 0.4 \cdot \text{Cost}$ :



What's wrong with this model?

# One-hot Encoding

To include a categorical variable in a regression model we need to represent it using numeric values. In the example below we map Type to the **dummy variable** named Type = "Public" using **one-hot encoding**:

| College                 | Type      |
|-------------------------|-----------|
| Grinnell College        | "Private" |
| University of Iowa      | "Public"  |
| University of Minnesota | "Public"  |
| Middlebury College      | "Private" |
| Carlton College         | "Private" |



| College                 | Type = "Public" |
|-------------------------|-----------------|
| Grinnell College        | 0               |
| University of Iowa      | 1               |
| University of Minnesota | 1               |
| Middlebury College      | 0               |
| Carlton College         | 0               |

Note that private colleges are the **reference group** in this example.

# One-hot Encoding

One-hot encoding is frequently used for nominal categorical variables with many categories:

| College                 | State |
|-------------------------|-------|
| Grinnell College        | IA    |
| University of Iowa      | IA    |
| University of Minnesota | MN    |
| Middlebury College      | VT    |
| Carlton College         | MN    |

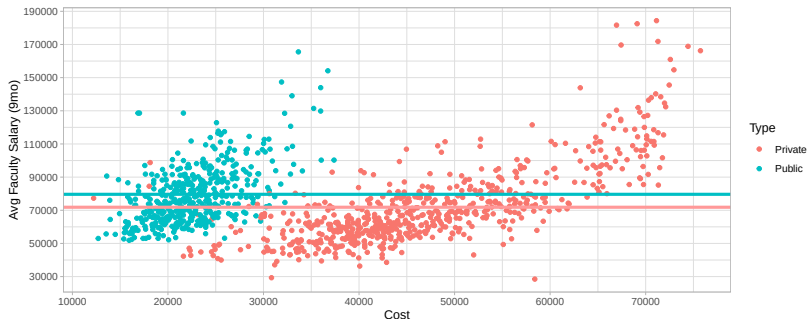


| College                 | State = "MN" | State = "VT" |
|-------------------------|--------------|--------------|
| Grinnell College        | 0            | 0            |
| University of Iowa      | 0            | 0            |
| University of Minnesota | 1            | 0            |
| Middlebury College      | 0            | 1            |
| Carlton College         | 1            | 0            |

Note that "IA" is the reference category in this example. We'll see why a reference category is necessary by looking at a few fitted models on our previous scatterplot.

## Example

Below is the fitted model  $\widehat{\text{Salary}} = 71836 + 7800 \cdot (\text{Type} = \text{"Public"})$ :

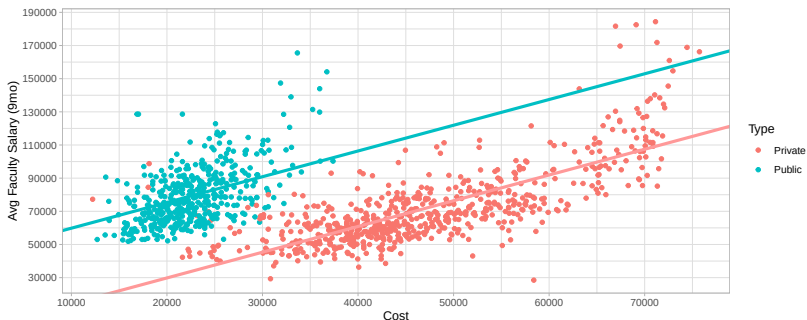


The model's intercept is the expected salary for private schools, illustrating why we use a reference group with one-hot encoding.

# Adjusted Effects

If we fit a model using *both* type and cost, we get:

$$\widehat{\text{Salary}} = -1229 + 45529 \cdot (\text{Type} = \text{"Public"}) + 1.55 \cdot \text{Cost}$$



How we interpret the effect of a college being “Public”? How does this differ from the previous model (that didn’t include cost)?

# Model Coefficient Interpretations

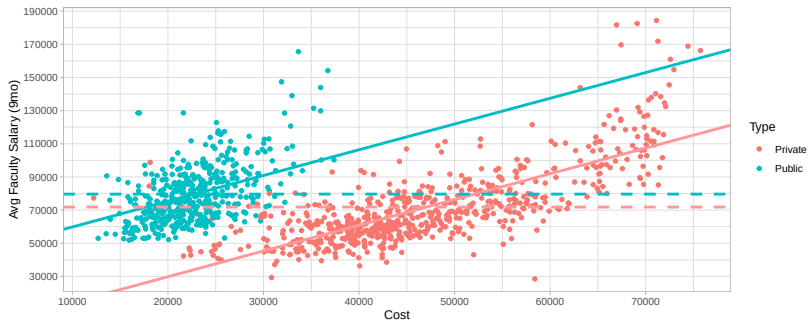
- ▶ Cost only:  $\widehat{\text{Salary}} = 60150 + 0.4 * \text{Cost}$ 
  - ▶ Averaging across both types (private and public), each \$1 increase in cost is expected to increase average faculty salary by \$0.4
- ▶ Cost and type:  
 $\widehat{\text{Salary}} = -1229 + 45529 * (\text{Type} = \text{'Public'}) + 1.55 * \text{Cost}$ 
  - ▶ For colleges of the same type, each \$1 increase in cost is expected to increase average faculty salary by \$1.55

## Model Coefficient Interpretations (cont.)

- ▶ Type only:  $\widehat{\text{Salary}} = 71836 + 7800 \cdot (\text{Type} = \text{'Public'})$ 
  - ▶ Averaging across colleges of all costs, faculty salaries are expected to be \$7800 higher for public colleges than private colleges
- ▶ Cost and type:  
 $\widehat{\text{Salary}} = -1229 + 45529 \cdot (\text{Type} = \text{'Public'}) + 1.55 \cdot \text{Cost}$ 
  - ▶ For colleges of the *same cost*, faculty salaries are on average \$45529 higher for public colleges than private colleges

# Is an Adjusted Effect Appropriate?

Suppose you're giving career advice to a recent PhD graduate. Is the marginal or adjusted effect of Type more useful in helping them decide which colleges they should apply to?

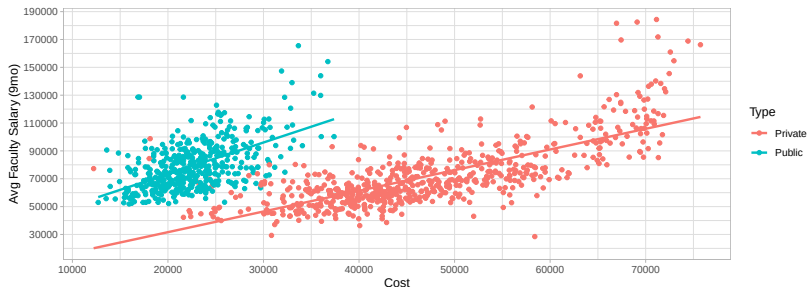


## Is an Adjusted Effect Appropriate? (cont.)

- ▶ The adjusted effect is *inappropriate* for this purpose, as it suggests that public colleges will pay better and should be favored in this individual's job search
  - ▶ In this scenario it isn't useful to adjust for cost, as the PhD graduate isn't concerned with differences in cost
  - ▶ This differs from our analysis of the Florida death penalty sentencing data where *did care* about adjusting for differences in the racial distribution of victims

# Stratification

The model  $\text{Avg\_Fac\_Salary} \sim \text{Type} + \text{Cost}$  forces a single slope coefficient for both private and public schools (which was sort of the point). We could allow for different slopes using *stratification*:



- ▶ Among private colleges:  $\widehat{\text{Salary}} = 1952.14 + 1.485 \cdot \text{Cost}$
- ▶ Among public colleges:  $\widehat{\text{Salary}} = 28025.86 + 2.267 \cdot \text{Cost}$

# Conclusion

- ▶ Categorical variables are represented in regression models via one-hot encoding
  - ▶ This designates one category as the reference group, and the estimated coefficients of dummy variables describe expected differences from this group
- ▶ Regression can be used to estimate *adjusted effects*, such as the estimated effect of cost within colleges of the same type, or the estimated difference between private and public colleges of the same cost
  - ▶ In some situations the adjusted effect is more meaningful than the marginal effect, but this isn't universally true, so we should avoid blindly adjusting for every variable present in our dataset